Ali Hajiabadi

Postdoctoral Researcher and Lecturer
ETH Zürich

Contact Information

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Research Interests

Systems Security, Formal Methods, Computer Architecture, Hardware/Software Co-design, Compilers, Secure Architectures and Software, Microarchitectural Side Channels, Machine Learning Systems Security

Education

2019 - 2024 Doctor of Philosophy in Computer Science
National University of Singapore (NUS), Singapore
Thesis: *"Building Efficient and Secure Processors thorugh Hardware/Software Co-design"*
Advisor: Prof. Trevor E. Carlson

2014 - 2019 Bachelor of Science in Computer Engineering
Sharif University of Technology, Tehran, Iran
Thesis: *"High Concurrency Latency Tolerant Register Files for GPUs"*
Advisor: Prof. Hamid Sarbazi-Azad

2009 - 2013 Diploma in Physics and Mathematics Discipline
Shahid Beheshti High School, Birjand, Iran
Affiliated with the National Organization for the Development of Exceptional Talents (NODET)

Honors & Awards

Jul. 2026	Noteworthy Reviewer, as a PC member of EuroS&P '26.
Aug. 2024	Dean's Graduate Research Excellence Award.
Jul. 2024	School of Computing CS PhD Thesis Award (Honorable Mention).
Jun. 2024	Best Paper Award nomination at DAC '24.
Oct. 2023	NUSGS Research Incentive Award from School of Computing, NUS (S$ 2,500 award).
Jan. 2022	Student Travel Award from ASPLOS'22 conference.
Aug. 2021	Research Achievement Award from School of Computing, NUS.
Mar. 2020	Invited talk and travel grant for the 2^{nd} Young Architect Workshop at ASPLOS'20, Switzerland.
Feb. 2019	President's Graduate Fellowship, the most prestigious doctoral fellowship in Singapore.
Sep. 2014	Ranked **164**th in Iranian National University Entrance Exam among more than 250,000 students.
2006/2009	Recognized as talented student in entry exam of NODET for middle school and high school.

Professional Experience

Jul. 2024 - Present
Postdoctoral Researcher at ETH Zürich, Switzerland
Computer Security (COMSEC) Group
Group Lead: Prof. Kaveh Razavi
Research: CPU and DRAM security, novel attacks and defenses for computing systems

Aug. 2019 - Jun. 2024
Graduate Research Assistant at National University of Singapore, Singapore
NUS Computer Architecture (CompArch) Group
Group Lead: Prof. Trevor E. Carlson
Research: hardware/software co-design for efficient and secure modern processors

Jul. 2016 - Jun. 2019
Research Assistant at Sharif University of Technology, Tehran, Iran
High Performance Computer Architectures and Networks (HPCAN) Lab
Group Lead: Prof. Hamid Sarbazi-Azad
Research: GPU register prefetching via HW/SW co-design and compiler optimizations

Summer 2018
Research Intern at National University of Singapore, Singapore
Group Lead: Prof. Trevor E. Carlson
Research: exploring new implementations for out-of-order commit processors

Peer-Reviewed Publications

17. CCS'26 — **Ali Hajiabadi**, Silvan Niederer, Kaveh Razavi
VeriPRAC: Verifiable Formal Contracts for Secure Per Row Activation Counting.
Proceedings of 33^{rd} ACM Conference on Computer and Communications Security (CCS 2026), November 2026.

16. DIMVA'26 — Silvan Niederer, **Ali Hajiabadi**, Kaveh Razavi
Glasswall: Enforcing User-Defined Data Usage Policies in Functions as a Service.
Proceedings of 23^{rd} Conference on Detection of Intrusions and Malware & Vulnerability Assessment, July 2026. Acceptance rate: 33/151 = 21.8%.

15. S&P'26 — Jean-Claude Graf*, Sandro Rüegge*, **Ali Hajiabadi**, Kaveh Razavi
VMScape: Exposing and Exploiting Incomplete Branch Predictor Isolation in Cloud Environments.
Proceedings of 47^{th} IEEE Symposium on Security and Privacy (S&P, Oakland 2026), May 2026. Acceptance rate: 118/925 = 12.8%.

14. HPCA'26 — Hritvik Taneja, **Ali Hajiabadi**, Michele Marazzi, Kaveh Razavi, Moinuddin K. Qureshi
MIRZA: Efficiently Mitigating Rowhammer with Randomization and ALERT.
Proceedings of 32^{nd} IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026), February 2026. Acceptance rate: 119/602 = 19.8%.

13. CCS'25 — **Ali Hajiabadi**, Michele Marazzi, Kaveh Razavi
CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation.
Proceedings of 32^{nd} ACM Conference on Computer and Communications Security (CCS 2025), October 2025. Acceptance rate: 316/2186 = 14.5%.

12. MICRO'25 — Silvan Niederer, Sandro Rüegge, **Ali Hajiabadi**, Kaveh Razavi
One Flew over the Stack Engine's Nest: Practical Microarchitectural Attacks on the Stack Engine.
Proceedings of 58^{th} IEEE/ACM International Conference on Microarchitecture (MICRO 2025), October 2025. Acceptance rate: 124/597 = 20.8%.

11. ISCA'25 — **Ali Hajiabadi**, Trevor E. Carlson
CASSANDRA*: Efficient Enforcement of Sequential Execution for Cryptographic Programs.*
Proceedings of 52^{nd} ACM/IEEE International Symposium on Computer Architecture (ISCA 2025), June 2025. Acceptance rate: 132/570 = 23.1%.

10. TACO'25 — Yun Chen*, **Ali Hajiabadi***, Romain Poussier, Yaswanth Tavva, Andreas Diavastos, Shivam Bhasin, Trevor E. Carlson
PARADISE*: Criticality-Aware Instruction Reordering for Power Attack Resistance.*
In ACM Transactions on Architecture and Code Generation (TACO), 2024.
*Joint first-authors with equal contribution.

9. DAC'24 — **Ali Hajiabadi**, Trevor E. Carlson
CONJURING*: Leaking Control Flow via Speculative Fetch Attacks.*
Proceedings of 61^{st} ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.
★ *Best Paper Award Nominee* (5/337 = 1.5%).

8. DAC'24 — **Ali Hajiabadi**, Archit Agarwal, Andreas Diavastos, Trevor E. Carlson
LEVIOSO*: Efficient Compiler-Informed Secure Speculation.*
Proceedings of 61^{st} ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.

7. HPCA'24 — Yun Chen*, **Ali Hajiabadi***, Trevor E. Carlson
GADGETSPINNER*: A New Transient Execution Primitive using the Loop Stream Detector.*
Proceedings of 30^{th} IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.
*Joint first-authors with equal contribution.

6. HPCA'24 — Yun Chen, **Ali Hajiabadi**, Lingfeng Pei, Trevor E. Carlson
PREFETCHX*: Cross-Core Cache-Agnostic Prefetcher-Based Side-Channel Attacks.*
Proceedings of 30^{th} IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.

5. ICCAD'23 — Arash Pashrashid, **Ali Hajiabadi**, Trevor E. Carlson
HIDFIX*: Efficient Mitigation of Cache-based Spectre Attacks through Hidden Rollbacks.*
Proceedings of 42^{nd} IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2023), November 2023. Acceptance rate: 172/768 = 22.4%.

4. **ICCAD'22** — Arash Pashrashid, **Ali Hajiabadi**, Trevor E. Carlson
Fast, Robust and Accurate Detection of Cache-based Spectre Attack Phases.
Proceedings of 41^{st} IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2022), November 2022. Acceptance rate: 132/586 = 22.5%.

3. **ASPLOS'21** — **Ali Hajiabadi**, Andreas Diavastos, Trevor E. Carlson
NOREBA*: A Compiler-Informed Non-speculative Out-of-Order Commit Processor.*
Proceedings of 26^{th} ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021), April 2021. Acceptance rate: 75/398 = 18.8%.

2. **TOCS'21** — Mohammad Sadrosadati, Amirhossein Mirhosseini, **Ali Hajiabadi**, Seyed Borna Ehsani, Hajar Falahati, Hamid Sarbazi-Azad, Mario Drumond, Babak Falsafi, Rachata Ausavarungnirun, Onur Mutlu
Highly Concurrent Latency-tolerant Register Files for GPUs.
In ACM Transactions on Computer Systems (TOCS), 2021.

1. **CGO'21** — Harish Patil, Alexander Isaev, Wim Heirman, Alen Sabu, **Ali Hajiabadi**, Trevor E. Carlson
ELFies: Executable Region Checkpoints for Performance Analysis and Simulation.
Proceedings of 19^{th} IEEE International Symposium on Code Generation and Optimization (CGO 2021), March 2021. Acceptance rate: 31/89 = 34.8%.

Teaching Experience

▶ **ETH Zürich, Switzerland**

Fall 2026	**Lecturer**, Computer Security (BSc) [**New: RISC-V edition**]
Spring 2026	**Co-lecturer**, Computer Engineering (BSc), *with Prof. Kaveh Razavi and Dr. Patrick Jattke*
Fall 2025	**Lecturer**, Computer Security (BSc) [**New Course**]
Spring 2025	**Lecturer**, Capture the Flag – Introduction to Cybersecurity (P&S BSc) [**New Course**]
Spring 2025	**Co-lecturer**, Computer Engineering (BSc), *with Prof. Kaveh Razavi*

▶ **National University of Singapore, Singapore**

Spring 2020 & Spring 2021	**Teaching Assistant** (*tutorial instructor*), CS2106 Introduction to Operating Systems (BSc) *Lecturer: Prof. Djordje Jevdjic*

▶ **Sharif University of Technology, Tehran, Iran**

Spring 2017	**Teaching Assistant** (*assignments & projects*), CE323 Computer Architecture (BSc) *Lecturer: Prof. Hamid Sarbazi-Azad*
Fall 2017 & Fall 2018	**Teaching Assistant** (*tutorial instructor, assignments & projects*), CE453 Real-Time Systems (BSc) *Lecturer: Prof. Amirhossein Jahangir*

Professional Services

Program Committee	S&P *('27)*, uASC *('27, '26)*, HPCA *('27, '26)*, MICRO *('26, '25)*, ISCA *('26)*, EuroS&P *('26)*, AsiaCCS *('26)*
Sub-reviewer	USENIX Security ('25)
Reviewer (journals)	IEEE CAL, ACM TACO, ACM TRETS
Session Chair	uASC'26 (*Brave New Defense*), MICRO'25 (*Security and Privacy - Memory*)
Shadow PC	EuroSys ('24, '23)
Student Volunteer	PLDI ('21)

Talks

Nov. 2025	**The Hidden and Ugly Faces of Spectre: Unexplored and Unmitigated Threat Models** *IC, EPFL*, Lausanne, Switzerland.
Oct. 2025	**CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation** *Conference on Computer and Communication Security (CCS 2025)*, Taipei, Taiwan.
Jun. 2025	CASSANDRA**: Efficient Enforcement of Sequential Execution for Cryptographic Programs** *Int. Symposium on Computer Architecture (ISCA 2025)*, Tokyo, Japan.
Mar. 2024	**Will CPUs Be Free of Spectre? Dark Side and Light Side of the Battle** *ETH Zürich, COMSEC Group*, Zürich, Switzerland.
Mar. 2024	GADGETSPINNER**: A New Transient Execution Primitive using the Loop Stream Detector** *Int. Symposium on High-Performance Computer Architecture (HPCA 2024)*, Edinburgh, Scotland.

Aug. 2021	Noreba: **A Compiler-Informed Non-speculative Out-of-Order Commit Processor** *Computing Research Week, School of Computing (NUS)*, Virtual.
Apr. 2021	Noreba: **A Compiler-Informed Non-speculative Out-of-Order Commit Processor** *Int. Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021)*, Virtual.
Feb. 2021	**Accelerating HPC applications with Out-of-Order Commit Processors** *Free and Open source Software Developers' European Meeting (FOSDEM 2021), HPC, Big Data, and Data Science track*, Virtual.
Mar. 2020	**Speculation-Free Out-of-Order Commit** *2^{nd} Young Architect Workshop at the 25^{th} International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)*, Virtual.

Research Mentoring

Mar. 2026 - Jun. 2026	*Yves Rüetschi*	BSc thesis, ETH Zürich	
Feb. 2026 - Jun. 2026	*Tomas Jossen*	BSc thesis, ETH Zürich	
Feb. 2026 - present	*Jonas Buchholz*	MSc thesis, ETH Zürich	*Embargo*
Jan. 2026 - present	*Quentin Bordier*	PhD thesis, COMSEC, ETH Zürich	*Embargo*
Mar. 2025 - May 2026	*Jean-Claude Graf*	PhD thesis, COMSEC, ETH Zürich	*1 publication (S&P'26)*
Mar. 2025 - May 2026	*Sandro Rüegge*	PhD thesis, COMSEC, ETH Zürich	*1 publication (S&P'26)*
Mar. 2025 - Sep. 2025	*Maximilian Mosler*	MSc thesis, ETH Zürich	*Embargo*
Mar. 2025 - Jun. 2025	*Jonas Buchholz*	Semester project, ETH Zürich	*Embargo*
Dec. 2024 - Jun. 2025	*Chenfei Liu*	MSc thesis, ETH Zürich	*1 publication (ASPLOS'27)*
Sep. 2024 - Feb. 2025	*Mounir Raki*	Semester project, ETH Zürich	
Jul. 2024 - present	*Silvan Niderer*	PhD thesis, COMSEC, ETH Zürich	*2 publications (MICRO'25, DIMVA'26)*
Jan. 2021 - Dec. 2023	*Arash Pashrashid*	PhD thesis, NUS CompArch	*2 publications (ICCAD'23, ICCAD'22)*
Jul. 2021 - Jul. 2023	*Archit Agarwal*	RA, NUS CompArch	*1 publication (DAC'24)*
Sep. 2020 - Sep. 2021	*Vernon Pang*	BSc final-year project, NUS	
Aug. 2020 - Mar. 2024	*Yun Chen*	PhD thesis, NUS CompArch	*3 publications (2$\times$HPCA'24, TACO)*

Skills

Programming Languages:	C, C++, Python, bash, and familiar with Java, Matlab, Scala
Instruction Set Architectures:	x86, Arm, RISC-V
Scientific Tools:	LLVM Compiler Infrastructure, gem5 Simulator, Sniper Simulator, Ramulator, Intel Pin, DynamoRIO